

Milestone 4

Display the result of a MPC query and final closeout

Project Catalyst #1200045: MPC as a Layer 2 Service to Cardano

Profila AG x Partisia Blockchain

Network: Cardano Preprod + Partisia Blockchain Testnet

Acceptance Criteria & Evidence

1. The result (dataset) of a MPC query will be documented and displayed to the initiator

MPC computation result: 5 (5 out of 5 submitted users have age > 18). Result fetched from live PBC contract state via REST API and displayed via terminal output and HTML evidence page. Aggregate result only - no individual data exposed.

MPC result:

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/relay/mpc-result.json

Result display script:

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/relay/show_result.ts

2. Documentation provided in PDF format in English

Link:

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/docs/evidence/milestone-4-report.pdf

3. Documentation includes the definition of the final process of displaying the result of a MPC query

The process: show_result.ts calls PBC REST API, receives serialized contract state, decodes ProfilaMpcState binary, extracts aggregate result, displays in terminal and generates HTML evidence page with cross-chain links.

E2E run log (full 6-stage pipeline):

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/docs/evidence/e2e-run-log.json

4. Video of the process uploaded to YouTube

M4 Demo:

<https://youtu.be/2YYvELDzrGo>

5. Video includes a demo of the end-to-end process as executed on the Cardano and PBC testnet, from quer

Cardano TX (query initiation):

<https://preprod.cardanoscan.io/transaction/50d28fd2bd263a84485b45280afd19bb4c3e20c24e9a4b52b6eac20418d53a57>

PBC Contract (result + state):

<https://browser.testnet.partisiablockchain.com/contracts/039ed9214602f2a93eb05411f852fa78a476607634>

Combined M1-M4 end-to-end demo:

<https://youtu.be/jwwfdTSUS2w>

6. Final closeout report is publicly available

Link:

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/docs/evidence/closeout-report.pdf

7. Final closeout video is publicly available

Link:

<https://youtu.be/jwwfdTSUS2w>

Evidence of Milestone Completion

Documentation (PDF):

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/docs/evidence/milestone-4-report.pdf

YouTube demo:

<https://youtu.be/2YYvELDzrGo>

Repository:

https://github.com/Profila/Aiken-Partisia_MPC

Stakeholder sign-off (Profila and Partisia):

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/docs/Profila%20%26%20Partisia%20Sign-off.pdf

Final closeout report:

https://github.com/Profila/Aiken-Partisia_MPC/blob/main/docs/evidence/closeout-report.pdf

Final closeout video:

<https://youtu.be/jwwfdTSUS2w>

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Project Catalyst Fund 13 - Proposal #1200045*